

1. AI Tool(s) Used

- ChatGPT
- Gemini

2. Scope of Usage

I wrote the majority of the HTML, CSS, and JavaScript logic for this project by myself. I only used AI for debugging and to learn specific API implementations that were new to me.

A. Firebase Real-time Synchronization

Location: js/chat.js (Message listener)

Prompt: "How do I use onSnapshot in Firebase to listen for changes without reloading the page?"

Response: AI provided a code snippet showing how to use `db.collection(...).onSnapshot()`.

Refinement: I studied the code and integrated it into my own chat rendering function so that the messages appear in the correct order. I also added my own logic to filter out messages from blocked users.

B. Gemini Bot Integration

Location: js/chat.js (Chatbot logic)

Prompt: "What is the correct fetch request format to send a message to the Gemini API using an API key?"

Response: AI provided the correct endpoint URL and the JSON structure required for the request body.

Refinement: I took the fetch logic and wrapped it inside my own `sendMessage` function, adding my own condition to only trigger the bot if the message text starts with "@bot".

C. CSS Flexbox Debugging

Location: style.css (Chat container)

Prompt: "My chatroom messages container is expanding beyond the screen and not scrolling. I'm using flexbox. How to fix?"

Response: AI suggested adding `min-height: 0` to the flex children and `overflow-y: auto` to the container.

Refinement: I applied these CSS rules to my classes to fix the scrolling issue on mobile devices.

3. Statement of Non-Usage

For the rest of the project, including the UI design, the sticker canvas logic, the GIF search, and the authentication flow, no AI tools were used. I wrote the code myself by referring to the official documentation and class materials.